-BIZI-

A WEB-BASED CIVIC-TECH AND BUSINESS NETWORKING PLATFORM

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DECLARATION

I declare that this or any other University has not previously submitted this work for the awarding of the course marks. I also declare that I will only present this project once the supervisor has approved and signed the document. To the best of my knowledge and belief, this work contains no material previously published or written by another person except where due reference is made.

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APPROVAL

This project proposal/ final project / dissertation of *student* was conducted under our supervision and is submitted with our approval as university supervisors.

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DEDICATION

I dedicate this project to Mr. and Mrs. Yator – my parents – for always supporting me in my journey to become a politician. Their advices on 'Peaceful Politics' has greatly influenced my project

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ABSTRACT

Kenya's Constitution (2010) guarantees public participation in legislative processes, yet a significant disconnect persists between the business community and Parliament. The Finance Bill 2024 crisis exemplified this gap, where inadequate awareness and fragmented feedback mechanisms contributed to nationwide protests that resulted in tragic loss of life. Business owners, particularly small and medium enterprises contributing 40% of Kenya's GDP, lack accessible platforms to track legislation, express opinions on proposed bills, and coordinate advocacy efforts. Existing parliamentary monitoring tools like Mzalendo provide informational resources but do not enable participatory engagement or business networking, leaving a critical void in Kenya's civic-tech landscape.

Bizi is a web-based civic-tech and business networking platform designed to address these challenges. The platform enables business owners, entrepreneurs, and economic enthusiasts to track parliamentary bills from introduction through presidential assent, vote on proposed legislation to express support or opposition, participate in moderated policy discussions, and build professional networks with like-minded stakeholders. Developed using the Agile Scrum methodology over a seven-week sprint cycle, Bizi employs React.js for the frontend, Node.js with Express.js for the backend, and PostgreSQL for data persistence. The system architecture prioritizes accessibility, security, and scalability while aligning with Mozilla's Responsible Computing principles of transparency, digital inclusion, and democratic empowerment.

The platform successfully implements user authentication with email verification, comprehensive bill tracking with stage progression visualization, a voting mechanism with real-time result aggregation and visual analytics, discussion forums with moderation capabilities, a connection-based networking system with private messaging, and an administrative dashboard for content management. Testing validates that the system meets functional requirements including vote integrity, notification delivery, and responsive design across devices. Bizi demonstrates that technology can bridge the gap between citizens and legislators, offering a model for civic engagement platforms that could prevent future crises by channeling public sentiment through constructive, democratic channels rather than confrontation.

Table of Contents

CHAPTER 1: Introduction 1
1.1 Background Information 1
1.2 Problem Statement 2
1.3 Objectives 3
1.4 Justification 3
1.5 Scope 9
1.6 Relationship Between the Project and Degree Programme 12
CHAPTER 2: LITERATURE REVIEW 14
2.1 Introduction 14
2.2 Civic Technology and Electronic Participation 14
2.2.1 Concept of Civic Technology 14
2.2.2 E-Participation Models and Frameworks 15
2.3.2 Existing Legislative Tracking Platforms 16
2.4.2 Features of Business Networking Platforms 16
2.5 Public Participation in Kenya's Legislative Process 17
2.5.1 Constitutional and Legal Framework 17
2.5.2 Challenges in Public Participation 18
2.6 Review of Existing Systems 19
2.6.1 Local Systems 19
2.6.2 International Systems 19
2.7 Technical Considerations 20
2.7.1 Web Development Frameworks 20
2.7.2 Database Management Systems 21
2.7.3 Authentication and Security 21
2.7.4 System Architecture Approaches 22
2.8 Identified Research Gaps 22

2.9 Conclusion 23
CHAPTER 3: METHODOLOGY 26
3.1 System Development Methodology 26
3.1.1 Overview of Development Methodologies 26
3.1.2 Selected Methodology: Agile with Scrum Framework 26
3.1.3 Scrum Framework Implementation 27
3.2 Requirements Gathering 28
3.2.1 Collection Method 28
3.2.2 Non-Functional Requirements 30
3.2.3 Functional Requirements 31
3.3 System Design 36
3.3.1 Block Diagram 36
3.3.2 Data Flow Diagram 39
3.3.3 Entity Relationship Diagram 41
3.3.4 Use Case Diagram 43
3.4 Technical Requirements 47
3.4.1 Development Tools and Technologies 47
3.4.2 Software Requirements 49
3.5 Budget 50
3.5.1 Budget Breakdown 50
3.6 Timeline 52
3.6.1 Sprint Schedule 52
3.6.3 Timeline Considerations 54

DEFINATION OF TERMS

- ✓ **API (Application Programming Interface):** The set of endpoints through which the Bizi frontend communicates with the backend server to manage data.
- ✓ **Authentication:** The process of verifying a user's identity through email and password credentials, issuing a JWT token upon success.
- ✓ **Bill:** A proposed piece of legislation under consideration by Kenya's Parliament, tracked on the Bizi platform.
- ✓ **Bill Category:** A classification (e.g., Taxation, Trade, Labor) assigned to bills for organizational purposes.
- ✓ **Bill Stage:** A specific phase in the legislative process including First Reading, Second Reading, Committee Stage, Third Reading, and Presidential Assent.
- ✓ **Bill Tracking:** The functionality enabling users to view, search, filter, and monitor parliamentary bills.
- ✓ **Bizi:** The web-based civic-tech and business networking platform developed in this project.
- ✓ **Business Community:** Business owners, entrepreneurs, and economic enthusiasts who are the primary target users of Bizi.
- ✓ **Business Networking:** Functionality enabling users to discover, connect with, and message other business professionals.
- ✓ **Civic Technology (Civic-Tech):** Digital tools that enhance citizen engagement with government processes and democratic decision-making.
- ✓ **Committee Stage:** The legislative phase where a bill is examined in detail by a parliamentary committee.
- ✓ **Connection:** A mutual relationship between two Bizi users enabling private messaging and enhanced visibility.
- ✓ **Connection Request:** A notification sent from one user to another expressing interest in establishing a professional connection.
- ✓ **Context Diagram:** A Level 0 Data Flow Diagram representing the system as a single process with external entities.
- ✓ **Dashboard:** The administrative interface providing tools for managing bills, users, and platform analytics.
- ✓ **Data Flow Diagram (DFD):** A graphical representation showing how data moves through the Bizi system.
- ✓ **Data Protection Act (2019):** Kenya's legislation governing personal data collection, processing, and storage.
- ✓ **Digital Inclusion:** The design principle ensuring platform accessibility regardless of technical expertise or device type.

- ✓ **Discussion Forum:** The feature enabling users to create threads and engage in policy-related conversations.
- ✓ **Discussion Post:** An individual reply or comment made within a discussion thread.
- ✓ **Discussion Thread:** A topic-based conversation initiated by a user, consisting of an original post and replies.
- ✓ **E-Participation:** The use of ICT to enable citizen engagement in governance processes.
- ✓ **Eco-Levy:** A proposed environmental tax from Finance Bill 2024, used as an example in this study.
- ✓ **Entity Relationship Diagram (ERD):** A graphical representation of the database structure showing tables and relationships.
- ✓ **Express.js:** The Node.js web framework used to build the Bizi backend API.
- ✓ **Finance Bill:** Annual legislation implementing tax measures, with Finance Bill 2024 serving as a case study.
- ✓ **First Reading:** The initial legislative stage where a bill is formally introduced in Parliament.
- ✓ **Follow (Bill):** The action of subscribing to receive notifications about a specific bill's updates.
- ✓ **Frontend:** The client-side component built with React.js that users interact with through browsers.
- ✓ **Functional Requirements:** Specifications defining what the platform must do, including specific features.
- ✓ **Gantt Chart:** A visual timeline showing the schedule of tasks across the development period.
- ✓ **Guest User:** An unauthenticated visitor who can view public content but cannot vote or post.
- ✓ **Internet Health:** Mozilla's concept describing the overall well-being of the internet as a global resource.
- ✓ **JSON Web Token (JWT):** A token format used to authenticate users and authorize access to protected resources.
- ✓ **Legislative Process:** The formal procedure through which bills are introduced, debated, and enacted in Parliament.
- ✓ **Legislator:** An elected Member of Parliament or Senator who participates in the legislative process.
- ✓ **Like:** A positive reaction users can give to discussion posts, indicating appreciation or agreement.
- ✓ **Minimum Viable Product (MVP):** The initial platform version with core features sufficient for user testing.
- ✓ **Mobile Responsive:** Design ensuring the interface adapts appropriately to different screen sizes.

- ✓ **Moderation:** Administrative review and enforcement of community guidelines on user content.
- ✓ **Mozilla Responsible Computing Challenge:** A Mozilla Foundation initiative encouraging ethical technology development.
- ✓ **MSMEs:** Micro, Small, and Medium Enterprises forming a significant portion of Kenya's economy.
- ✓ **Networking Module:** The component handling user discovery, connection management, and private messaging.
- ✓ **Node.js:** The JavaScript runtime environment used to build the backend server.
- ✓ **Non-Functional Requirements:** Specifications defining quality attributes such as performance, security, and usability.
- ✓ **Notification:** An alert informing users of relevant events such as bill updates or new messages.
- ✓ **Notification Preferences:** User-configurable settings determining which notifications are received.
- ✓ **Parliamentary Bill:** A formal proposal for new legislation introduced in Kenya's Parliament.
- ✓ **Parliamentary Monitoring Organization (PMO):** A civil society entity that observes and reports on legislative activities.

LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
AWS	Amazon Web Services
C2C	Citizen-to-Citizen
C2G	Citizen-to-Government
CRUD	Create, Read, Update, Delete
CSRF	Cross-Site Request Forgery
CSS	Cascading Style Sheets
DB	Database
DFD	Data Flow Diagram
DPA	Data Protection Act
DPI	Dots Per Inch
ERD	Entity Relationship Diagram
FK	Foreign Key
FYP	Final Year Project
GB	Gigabyte
GDP	Gross Domestic Product
GUI	Graphical User Interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IAP2	International Association for Public Participation
ICT	Information and Communication Technology
IDE	Integrated Development Environment
I/O	Input/Output
JWT	JSON Web Token
KES	Kenya Shillings
KNCCI	Kenya National Chamber of Commerce and Industry
LAMP	Linux, Apache, MySQL, PHP
LTS	Long Term Support
Mbps	Megabits Per Second
MERN	MongoDB, Express.js, React.js, Node.js
MP	Member of Parliament
MS	Microsoft
MSMEs	Micro, Small, and Medium Enterprises
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
NoSQL	Not Only SQL

npm	Node Package Manager
ORM	Object-Relational Mapping
OS	Operating System
OWASP	Open Web Application Security Project
PK	Primary Key
PMO	Parliamentary Monitoring Organization
PNG	Portable Network Graphics
px	Pixels
RAD	Rapid Application Development
RAM	Random Access Memory
RDBMS	Relational Database Management System
REST	Representational State Transfer
SACCO	Savings and Credit Cooperative Organization
SME	Small and Medium Enterprise
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
SSD	Solid State Drive
SSL	Secure Sockets Layer
SVG	Scalable Vector Graphics
TLS	Transport Layer Security
UAT	User Acceptance Testing
UI	User Interface
UK	Unique Key
URL	Uniform Resource Locator
UX	User Experience
VAT	Value Added Tax
vCPU	Virtual Central Processing Unit

CHAPTER 1: Introduction

In today's rapidly evolving digital landscape, the intersection of technology, business, and governance presents unprecedented opportunities for civic engagement. As Kenya continues to strengthen its democratic institutions, there remains a significant gap between the business community and the legislative processes that directly impact economic activities. This chapter introduces Bizi, a civic-tech platform designed to empower business owners, entrepreneurs, and economic enthusiasts to actively participate in Kenya's legislative process while fostering meaningful business networking opportunities.

1.1 Background Information

Kenya's legislative framework plays a pivotal role in shaping the business environment within the country. Parliament enacts numerous bills annually that directly affect taxation, trade regulations, business licensing, labor laws, and investment policies. According to the Kenya National Bureau of Statistics (2023), Micro, Small, and Medium Enterprises (MSMEs) contribute approximately 40% of Kenya's Gross Domestic Product (GDP) and employ over 14 million Kenyans. Despite this significant economic contribution, the voice of the business community in the legislative process remains remarkably limited.

The Constitution of Kenya (2010) under Article 118 guarantees public participation in parliamentary proceedings. Citizens have the right to petition Parliament, participate in public hearings, and contribute to the legislative process. However, the practical implementation of these provisions has faced numerous challenges. Many business owners lack awareness of proposed bills, struggle to access parliamentary information, and find it difficult to navigate the complex legislative procedures.

The digital revolution has transformed how citizens interact with governments worldwide. E-participation platforms have emerged as powerful tools for enhancing civic engagement and democratic participation. Countries such as Estonia, South Korea, and Brazil have successfully implemented digital platforms that allow citizens to participate in governance processes. In Africa, similar initiatives are gaining momentum, with platforms like Mzalendo in Kenya tracking parliamentary activities.

Furthermore, the rise of social networking and digital business communities has demonstrated the power of connected entrepreneurs. Platforms such as LinkedIn, industry-specific forums, and business associations have shown that when business people connect, share ideas, and

collaborate, innovation flourishes and economic growth accelerates. However, no single platform currently combines legislative participation with business networking in the Kenyan context.

Bizi emerges from this context as a comprehensive solution that bridges the gap between Kenya's business community and the legislative process while simultaneously providing a robust networking platform for entrepreneurs and economic enthusiasts.

1.2 Problem Statement

Despite constitutional provisions guaranteeing public participation in legislative processes, Kenya's business community faces significant barriers to meaningful engagement with Parliament. The following critical issues necessitate the development of the Bizi platform:

Limited Awareness of Legislative Activities: A majority of business owners and entrepreneurs remain uninformed about bills under consideration in Parliament until they become law. There is no centralized, user-friendly platform that specifically targets the business community to provide real-time updates on economic and business-related legislation.

Complexity of Legislative Tracking: The Kenyan legislative process involves multiple stages including first reading, second reading, committee stage, third reading, and presidential assent. For the average business person, understanding and tracking these stages is complicated and time-consuming. Existing parliamentary resources are often technical and not designed for easy public consumption.

Absence of Business-Centric Voting Mechanisms: While platforms exist for general civic participation, there is no dedicated space where business stakeholders can collectively express their views on proposed legislation through structured voting and feedback mechanisms. This absence means that the collective voice of the business community is fragmented and often unheard.

Fragmented Business Networking: Entrepreneurs and business owners in Kenya often operate in silos, missing opportunities for collaboration, mentorship, and knowledge sharing. Existing social platforms are not tailored to address the specific needs of the Kenyan business community, particularly in relation to policy discussions and economic matters.

Disconnect Between Policy and Business Realities: Policymakers often lack direct feedback channels from those who will be most affected by their decisions. This disconnect can lead to the enactment of laws that, while well-intentioned, may have unintended negative consequences on businesses and economic growth.

These problems collectively result in a business community that is reactive rather than proactive in matters of policy, leading to missed opportunities for influence and, frequently, economic losses when unfavourable legislation is enacted.

1.3 Objectives

1.3.1 General Objectives

The general objective of this project is to design and develop Bizi, a web-based civic-tech and business networking platform that enables business owners, entrepreneurs, and economic enthusiasts in Kenya to actively participate in the legislative process, track parliamentary bills, express their views through structured voting mechanisms, and network with like-minded individuals for business growth and collaboration.

1.3.2 Specific Objectives

The specific objectives of this project are:

i. To design and implement a bill tracking module that retrieves, categorises, and displays information about Parliamentary bills relevant to business and the economy, allowing users to monitor legislation from introduction through presidential assent.

ii. To develop a voting and feedback mechanism that enables registered users to express their opinions on proposed bills, aggregate community sentiments, and generate reports that can be shared with relevant stakeholders, including legislators.

iii. To create a networking and discussion forum that facilitates both public and private interactions among business owners and entrepreneurs, enabling knowledge sharing, collaboration opportunities, and policy discussions.

iv. To implement a user authentication and profile management system that verifies users, maintains their preferences, tracks their participation history, and enables personalized content delivery based on business interests and industry categories.

1.4 Justification

The development of Bizi is justified by several compelling factors that underscore its potential impact on Kenya's business ecosystem and democratic processes:

Enhancing Democratic Participation: By providing an accessible platform for business stakeholders to engage with the legislative process, Bizi directly supports the constitutional mandate for public participation. The platform democratizes access to parliamentary information and empowers citizens to fulfill their civic responsibilities.

Economic Impact: When business owners are informed about and can influence legislation, they are better positioned to plan, adapt, and thrive. This proactive engagement can prevent

economic losses that result from sudden policy changes and can contribute to a more business-friendly legislative environment.

Aggregating Business Voices: Individual business owners often lack the resources or influence to be heard by policymakers. Bizi aggregates these voices, creating a collective platform where the combined sentiment of the business community carries weight. This collective voice can serve as a valuable input for legislators seeking to understand the practical implications of proposed laws.

Fostering Business Networks: Beyond legislative participation, Bizi addresses the need for meaningful business connections. By bringing together entrepreneurs, mentors, investors, and business enthusiasts, the platform creates opportunities for collaboration, partnership, and growth that extend beyond policy matters.

Bridging Information Gaps: Bizi translates complex legislative processes into understandable content, making parliamentary activities accessible to users regardless of their legal or political expertise. This educational component strengthens overall civic literacy within the business community.

Leveraging Technology for Governance: The platform aligns with Kenya's Vision 2030 and the Digital Economy Blueprint, which emphasize the use of technology to enhance governance and economic growth. Bizi represents a practical application of these national aspirations.

Case Study: Bizi and the June 2024 Finance Bill Crisis

Hypothetical Impact Analysis: Bizi During the Finance Bill 2024 Protests

The Finance Bill 2024 Crisis

In June 2024, Kenya witnessed unprecedented public unrest triggered by the Finance Bill 2024. The proposed legislation contained numerous tax measures that were perceived as burdensome to ordinary citizens and businesses alike. What began as online criticism rapidly escalated into nationwide protests, predominantly led by young Kenyans who mobilised under the banner of "Reject Finance Bill 2024."

The key provisions that sparked outrage included:

Proposed Measure	**Affected Stakeholders**
16% VAT on bread	Consumers, bakeries, retailers
Eco-levy on manufactured goods	Manufacturers, importers, retailers
Motor vehicle tax (2.5% annually)	Vehicle owners, transport businesses
25% excise duty on vegetable oils	Food manufacturers, households
Digital content creation tax	Content creators, digital entrepreneurs
Mobile money transfer tax increase	All mobile money users, small traders

VAT on financial services	Banks, SACCOs, insurance companies
Excise duty on telephone and data services	Telecom companies, all mobile users

The protests, which peaked on June 25, 2024, resulted in the tragic loss of life when demonstrators breached Parliament buildings and security forces responded with force. Dozens of Kenyans, predominantly young people, lost their lives. The crisis ultimately led President William Ruto to decline assent to the bill on June 26, 2024, marking a significant moment in Kenya's democratic history.

1.9.2 Analysis of Systemic Failures

The Finance Bill 2024 crisis exposed critical failures in Kenya's public participation infrastructure:

Failure 1: Inadequate Early Warning Systems

The Finance Bill was published and moved through initial parliamentary stages with limited public awareness. By the time citizens understood the full implications of the proposed taxes, the bill had already progressed significantly through the legislative process. Business owners, in particular, were caught off guard by provisions that would directly impact their operations.

Failure 2: Fragmented Business Community Response

The business community's response was disjointed and delayed. While individual business associations issued statements, there was no unified platform for aggregating business sector concerns or coordinating advocacy efforts. Small and medium enterprises, which would bear significant burden from the proposed taxes, lacked organised channels to voice their concerns.

Failure 3: Information Asymmetry

Complex tax provisions were buried in technical legislative language that ordinary citizens and business owners struggled to understand. The implications of various clauses only became clear through piecemeal social media explanations, sometimes accompanied by misinformation or exaggeration.

Failure 4: Absence of Constructive Dialogue Channels

The discourse around the Finance Bill became increasingly polarized on social media platforms. Twitter (X) and TikTok, while effective for mobilization, were not designed for nuanced policy discussion. The conversation devolved into hashtag activism without structured feedback mechanisms that could inform legislative review.

Failure 5: Disconnect Between Citizens and Legislators

Members of Parliament who voted for the bill faced intense public backlash, with many claiming they were unaware of the depth of public opposition. There was no mechanism for legislators to gauge real-time constituent sentiment on specific bill provisions.

Failure 6: Escalation to Physical Confrontation

The absence of effective digital participation channels contributed to the perception that physical protest was the only viable means of being heard. The tragic events of June 25, 2024, might have been avoided if structured platforms for engagement had existed.

1.9.3 How Bizi Could Have Made a Difference

Had the Bizi platform been operational during the Finance Bill 2024 process, it could have intervened at multiple points to facilitate constructive engagement and potentially avert the crisis.

Phase 1: Early Awareness and Mobilisation (April - May 2024)

Bill Tracking and Alerts

When the Finance Bill 2024 was first published and introduced in Parliament, Bizi would have:

- **Immediately categorised** the bill under relevant business categories (Taxation, Trade, Finance)
- **Sent automated notifications** to users who had expressed interest in tax-related legislation
- **Displayed the bill prominently** on the platform with its current stage clearly indicated

Simplified Content Delivery

The platform would have provided:

- **Plain-language summaries** of each tax provision
- **Business impact assessments** explaining how each measure would affect different sectors
- **Visual breakdowns** of proposed tax rates compared to existing rates

Phase 2: Structured Feedback Collection (May - Early June 2024)

Voting Mechanism in Action

As users became aware of the bill's provisions, Bizi's voting mechanism would have enabled:

- **Provision-by-provision voting:** Users could vote Support, Oppose, or Neutral on individual clauses
- **Sector-specific aggregation:** Results could be filtered by industry to show, for example, how transport sector businesses viewed the motor vehicle tax
- **Comment collection:** Optional comments would capture specific concerns and suggestions

Aggregated Reports for Legislators

Bizi would have generated comprehensive reports showing:

- Overall business community sentiment on the bill
- Breakdown by sector and enterprise size
- Specific concerns and suggested amendments
- Geographic distribution of responses

These reports could have been formally submitted to:

- The Departmental Committee on Finance and National Planning
- The National Treasury
- Individual Members of Parliament
- The Kenya National Chamber of Commerce and Industry

Phase 3: Constructive Dialogue Facilitation (June 2024)

Discussion Forums as an Alternative to Social Media

While Twitter and TikTok became battlegrounds of outrage, Bizi's discussion forums could have provided:

- **Moderated conversations** focused on specific provisions and their implications
- **Expert contributions** from economists, tax consultants, and policy analysts
- **Constructive proposals** for amendments rather than outright rejection
- **Cross-sector dialogue** connecting businesses affected by different provisions

Sample Discussion Thread on Bizi:

Thread: Alternative Approaches to Eco-Levy Implementation

Posted by: Manufacturing Association Representative

"While we understand the environmental objectives, the current eco-levy proposal would increase our production costs by 12%. We propose a phased implementation over 3 years with exemptions for businesses that adopt certified green practices..."

Replies: 234 | Likes: 1,892 | Views: 15,670

Reducing Misinformation

The platform's verified bill information would have:

- Provided authoritative source material for discussions
- Countered false claims about bill provisions
- Clarified which provisions were actual proposals versus rumours
- Distinguished between the National Assembly and public proposals

Scenario: Digital Content Creator Tax

What Actually Happened:
Young content creators, many of whom organised the protests, learned of the proposed tax on digital content through viral TikTok videos. The response was immediate outrage, but lacked structured advocacy.

With Bizi:

- Digital entrepreneurs registered on the platform would have been notified under the "Digital Economy" category
- Voting data would have shown overwhelming opposition from the tech and creative sectors
- Discussion forums would have enabled content creators to articulate specific concerns about Kenya's digital economy competitiveness
- Coalition building with tech companies and digital agencies would have amplified the advocacy

- Alternative taxation models could have been proposed through structured feedback

The Role of Constructive Engagement in Preventing Violence

The tragic violence of June 25, 2024, represented a failure of democratic engagement mechanisms. When citizens feel unheard through formal channels, the likelihood of resorting to physical confrontation increases.

1.5 Scope

The scope of the Bizi platform is defined by the following parameters:

Functional Scope:

- The platform will focus on Kenyan Parliamentary bills, specifically those with direct or indirect implications on business, trade, investment, taxation, and economic activities.
- Users will be able to register, create profiles, and specify their business interests and industry categories.
- The voting functionality will allow users to indicate support, opposition, or neutrality on bills, with optional comments.
- Discussion forums will support both public threads visible to all users and private messaging between connected users.
- Bill tracking will cover the stages from introduction in Parliament through presidential assent or rejection.
- The platform will provide notifications to users based on their interests and tracked bills.

Technical Scope:

- The system will be developed as a web-based application accessible through standard web browsers.
- The initial version will not include a native mobile application, though the web interface will be responsive and mobile-friendly.
- Bill information will be sourced from publicly available parliamentary records and may require manual updates where automated integration is not feasible.

Geographic Scope:

- The platform targets users within Kenya and focuses exclusively on the Kenyan Parliament (National Assembly and Senate).

- County-level legislation and county assembly activities are outside the scope of this initial version.

User Scope:

- The primary users are business owners, entrepreneurs, and individuals with an interest in economic and business matters.
- While the platform is open to all Kenyan citizens, its content and features are tailored specifically to business-oriented users.

Time Scope:

- The project will be developed within the academic timeframe allocated for the final year project.
- Features will be prioritized, with core functionalities (bill tracking, voting, user management) taking precedence over advanced features (analytics dashboards, API integrations).

Limitations:

- The platform does not replace formal parliamentary submission processes; it serves as a complementary tool for awareness and collective expression.
- Real-time integration with official parliamentary systems depends on the availability and accessibility of such data sources.
- The platform will not provide legal advice.

1.5.1 Relationship of Bizi to Mozilla's responsible computing

Promoting Civic Engagement and Democratic Participation

At the heart of Mozilla's vision is the belief that technology should empower citizens to participate meaningfully in democratic processes. The internet should serve as a tool for civic engagement, enabling individuals to access information, voice their opinions, and influence decisions that affect their lives.

The Bizi platform directly embodies this principle by creating a dedicated space for Kenya's business community to engage with the legislative process. By enabling users to track parliamentary bills, vote on proposed legislation, and participate in discussions on policy matters, Bizi transforms passive citizens into active participants in governance. The platform

operationalises the constitutional right to public participation (Article 118, Constitution of Kenya 2010) through accessible digital tools, thereby strengthening democratic processes.

The voting mechanism within Bizi aggregates the collective voice of the business community, creating a measurable expression of stakeholder sentiment that can inform legislators and policymakers. This approach aligns with Mozilla's advocacy for technology that enhances rather than undermines democratic institutions.

Ensuring Digital Inclusion and Accessibility

Mozilla emphasises that technology should be designed to include, not exclude. Digital solutions should be accessible to users regardless of their technical expertise, economic status, geographic location, or physical abilities. Responsible computing means creating products that bridge digital divides rather than widen them.

Bizi addresses digital inclusion through several design decisions:

- **Responsive Design:** The platform is built to function seamlessly across devices, from desktop computers to mobile phones. Given the high mobile phone penetration in Kenya, this ensures that business owners can access the platform regardless of the device they own.
- **Simplified Content:** Parliamentary information is often presented in technical legal language that can be inaccessible to non-experts. Bizi includes simplified summaries of bills that highlight business implications in plain language, making legislative content accessible to users without legal or political backgrounds.
- **Low Bandwidth Consideration:** Recognising that internet connectivity in Kenya varies in quality and cost, the platform is designed to be efficient, minimising data usage while maintaining functionality.

Fostering Transparency and Accountability

Mozilla advocates for technology that is transparent in its operation and accountable for its impacts. Users should understand how platforms work, how decisions are made, and how their data and contributions are used.

Bizi demonstrates transparency and accountability through:

- **Open Voting Results:** Vote tallies on bills are displayed publicly, allowing users to see how the community has responded to proposed legislation. This transparency builds trust and encourages participation.

- **Algorithmic Transparency:** Where recommendations are made (such as suggested bills or connections), the platform uses straightforward criteria (industry matching, stated interests) rather than opaque algorithms that users cannot understand.
- **Audit Trails:** Administrative actions (content moderation, user management) are logged, creating accountability for platform governance decisions.

Addressing Power Imbalances

Mozilla recognises that technology can either reinforce or challenge existing power imbalances. Responsible computing involves designing systems that empower marginalised or underrepresented groups rather than concentrating power among those who already hold it.

Bizi addresses power imbalances in the legislative process:

- **Amplifying Underrepresented Voices:** Small business owners and individual entrepreneurs often lack the resources or connections to participate in formal legislative processes. Bizi provides them with a platform where their voices can be aggregated and amplified alongside larger businesses and organized associations.
- **Democratizing Information Access:** By providing accessible, timely information about parliamentary bills, Bizi reduces the information asymmetry that often benefits well-resourced organizations at the expense of smaller players.
- **Collective Action:** The networking and discussion features enable business owners to organize, share perspectives, and potentially form coalitions around policy positions, enhancing their collective influence.

1.6 Relationship Between the Project and Degree Programme

This project demonstrates a strong alignment with the core competencies and learning outcomes of my degree programme in Computer Science. The development of Bizi integrates multiple domains of knowledge acquired throughout my academic journey:

Software Development : The project applies programming concepts, database management, web development frameworks, and software engineering principles learned across various courses. Designing a full-stack web application exercises both front-end and back-end development competencies.

Systems Analysis and Design: The requirement gathering, system modeling, and design phases of this project directly apply methodologies taught in systems analysis courses. Creating use

cases, data models, and architectural designs reinforces theoretical knowledge with practical application.

Database Management: Implementing the data layer of Bizi requires competent application of database design principles, normalization, query optimization, and data integrity concepts.

User Experience Design: Creating an intuitive and accessible platform for diverse users applies principles of human-computer interaction and user-centered design.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of literature relevant to the development of the Bizi platform. The review examines existing scholarly work, technical documentation, and practical implementations in areas that intersect with the project's objectives. The discussion is organized thematically to provide a coherent understanding of the concepts, technologies, and systems that inform the design and development of Bizi.

2.2 Civic Technology and Electronic Participation

2.2.1 Concept of Civic Technology

Civic technology, commonly referred to as civic-tech, encompasses the use of digital tools and platforms to enhance the relationship between citizens and their governments. According to Knight Foundation (2013), civic technology can be defined as "***technology that enables engagement or participation of the public for stronger development, enhancing citizen communications, improving government infrastructure, and generally improving the public good.***" This definition captures the broad spectrum of applications that fall under the civic-tech umbrella, ranging from simple information portals to sophisticated participatory platforms.

Patel et al. (2013) categorize civic technology into two primary domains: citizen-to-citizen (C2C) and citizen-to-government (C2G) applications. C2C applications focus on connecting citizens with each other to address community issues, share resources, and collaborate on local initiatives. C2G applications, on the other hand, facilitate interactions between citizens and government institutions, enabling participation in democratic processes, access to public services, and oversight of government activities. The Bizi platform incorporates elements of both categories, as it connects business owners with each other (C2C) while also facilitating their engagement with parliamentary processes (C2G).

Steinberg (2014) argues that effective civic technology must go beyond mere information provision to enable meaningful participation. The author identifies three levels of civic engagement: informational (one-way communication from government to citizens), consultative (government seeking input from citizens), and participatory (citizens actively involved in decision-making). Platforms that achieve the participatory level demonstrate greater impact on democratic processes and citizen satisfaction. This framework informs the design of Bizi, which aims to move beyond informational and consultative engagement to enable active participation through voting mechanisms and collective voice aggregation.

The emergence of Web 2.0 technologies has significantly accelerated the development of civic-tech platforms. O'Reilly (2005) describes Web 2.0 as "the network as platform," emphasizing user-generated content, collaboration, and interactivity. These principles have been adopted in civic technology to create platforms where citizens are not passive consumers of information but active contributors to democratic discourse. Social media integration, commenting systems, and collaborative features have become standard components of modern civic-tech solutions (Mergel, 2013).

2.2.2 E-Participation Models and Frameworks

Electronic participation (e-participation) refers to the use of information and communication technologies (ICT) to broaden and deepen political participation by enabling citizens to connect with each other and with their elected representatives (Macintosh, 2004). The United Nations E-Government Survey (2020) identifies e-participation as a critical indicator of digital governance maturity, measuring the extent to which governments provide online tools for citizen engagement.

Macintosh (2004) proposes a comprehensive framework for evaluating e-participation initiatives, identifying ten key dimensions: accessible technology, participation methods, information quality, usability, trust, political culture, citizen motivation, government motivation, evaluation, and sustainability. This framework provides a useful lens for analyzing existing platforms and designing new ones. For the Bizi platform, particular attention must be paid to accessibility, usability, and citizen motivation, as these factors directly influence adoption and sustained engagement among business users.

The International Association for Public Participation (IAP2) developed the Spectrum of Public Participation, which outlines five levels of engagement: inform, consult, involve, collaborate, and empower (IAP2, 2018). Each level represents increasing degrees of public impact on decisions. Table 2.1 illustrates this spectrum and its application to the Bizi platform.

Table 2.1: IAP2 Spectrum of Public Participation Applied to Bizi

Level	Description	Bizi Application
Inform	Providing balanced and objective information	Bill tracking and legislative updates
Consult	Obtaining feedback on alternatives and decisions	Voting mechanism on proposed bills
Involve	Working with the public throughout the process	Discussion forums and ongoing engagement
Collaborate	Partnering in each aspect of decision-making	Aggregated reports shared with legislators
Empower	Placing final decision-making in public hands	Outside platform scope; requires policy change

2.3.2 Existing Legislative Tracking Platforms

Several platforms have been developed globally to track legislative activities and facilitate public engagement with parliaments. These systems vary in their scope, features, and target audiences.

GovTrack (United States): Launched in 2004, GovTrack.us is one of the pioneering legislative tracking platforms (Tauberer, 2012). The platform provides comprehensive information on bills, votes, and members of the United States Congress. Key features include bill status tracking, legislator profiles, voting records, and email alerts for followed legislation. GovTrack demonstrates the value of presenting complex legislative information in accessible formats and has influenced the development of similar platforms worldwide.

TheyWorkForYou (United Kingdom): Developed by mySociety, TheyWorkForYou.com monitors the UK Parliament and devolved assemblies (mySociety, 2018). The platform allows users to track debates, search Hansard records, and monitor their representatives' activities. A notable feature is the ability to set up email alerts for specific topics or keywords mentioned in parliamentary proceedings. The platform has been credited with increasing citizen awareness of parliamentary activities and has served as a model for similar initiatives in other countries.

OpenParlamento (Italy): This platform tracks the Italian Parliament, providing information on bills, votes, and legislator activities (OpenPolis Association, 2015). OpenParlamento incorporates social features that allow users to comment on legislation and engage with other citizens. The platform also generates legislator "report cards" that summarize attendance, voting patterns, and legislative productivity.

Mzalendo (Kenya): Mzalendo.com is Kenya's primary parliamentary monitoring platform, operated by Mzalendo Trust (Mzalendo Trust, 2017). The platform provides information on Members of Parliament, including their profiles, attendance records, and contributions to debates. Users can search for bills and track their progress through Parliament. Ho

2.4.2 Features of Business Networking Platforms

Modern business networking platforms incorporate a range of features designed to facilitate professional relationship-building and knowledge sharing. A review of leading platforms reveals common patterns and innovative approaches.

Profile Systems: User profiles serve as digital representations of professional identity. Effective profile systems allow users to showcase their skills, experience, education, and accomplishments (Van Dijck, 2013). For business-focused platforms, profiles typically include business information such as company name, industry, size, and offerings. Bizi's profile system

should capture not only personal professional information but also business interests and industry categories relevant to legislative tracking.

Connection Mechanisms: Platforms employ various approaches to facilitate connections between users. LinkedIn uses a connection request model requiring mutual acceptance, while Twitter employs a follow model allowing one-way relationships (Kwak et al., 2010). Hybrid approaches are also common, with public following for content visibility and private connections for direct messaging. Bizi should implement a connection system that supports both public interactions (discussion forums) and private networking (direct messaging).

Content Sharing and Discussion: The ability to share content and participate in discussions is central to engagement on networking platforms. Kietzmann et al. (2011) identify seven functional building blocks of social media: identity, conversations, sharing, presence, relationships, reputation, and groups. Discussion forums and group features enable topical conversations that bring together users with shared interests. For Bizi, discussions centered on specific bills or economic topics create natural focal points for engagement.

Recommendation and Discovery: Algorithms that recommend connections, content, and opportunities enhance user experience and platform stickiness. Collaborative filtering and content-based recommendation systems help users discover relevant connections and information (Linden et al., 2003). Bizi can leverage user-specified interests and industry categories to recommend relevant bills for tracking and suggest connections with users in similar industries.

Privacy and Access Control: Business networking platforms must provide granular privacy controls, as users have varying comfort levels with information sharing (Stutzman et al., 2012). Features such as private messaging, closed groups, and visibility settings empower users to manage their digital presence. Bizi should offer clear distinctions between public forums accessible to all users and private networking features limited to connected users.

2.5 Public Participation in Kenya's Legislative Process

2.5.1 Constitutional and Legal Framework

The Constitution of Kenya (2010) establishes a robust framework for public participation in governance. Article 1 vests sovereign power in the people of Kenya, establishing the foundational principle that citizens should be involved in decision-making processes. Article 10 identifies public participation as one of the national values and principles of governance that binds all state organs, state officers, public officers, and all persons in the application and interpretation of the Constitution.

Article 118 specifically addresses public access and participation in Parliament. It states that Parliament shall conduct its business in an open manner and shall facilitate public participation and involvement in the legislative and other business of Parliament. Furthermore, Parliament is required to hold its sittings and those of its committees in public, though exceptions exist for specific circumstances outlined in legislation.

The County Governments Act (2012) and the Public Finance Management Act (2012) further reinforce public participation requirements at both national and county levels. These laws mandate that citizens be given opportunities to participate in budgetary processes, development planning, and policy formulation (Bosire, 2013).

Ghai (2014) observes that Kenya's 2010 Constitution represents one of the most participatory constitutional frameworks in Africa. The author notes that the constitution was itself a product of extensive public participation, with citizens contributing to the drafting process through thematic committees and national dialogue. This participatory ethos extends to the implementation of the constitution, with courts interpreting participation requirements broadly.

2.5.2 Challenges in Public Participation

Despite the strong constitutional framework, the implementation of public participation in Kenya faces significant challenges. Muriu (2014) identifies several barriers including limited awareness among citizens, inadequate resources for participation mechanisms, elite capture of participatory spaces, and insufficient feedback to participants.

A study by the International Budget Partnership (2019) found that while Kenya has made progress in providing budget information, opportunities for public participation in budget processes remain limited in practice. Many citizens lack awareness of their participation rights and the mechanisms available for engagement. This finding is consistent with the problem statement of the Bizi project, which notes that business owners often remain uninformed about proposed legislation.

Transparency International Kenya (2018) conducted an assessment of public participation in the legislative process and found that existing mechanisms primarily benefit organized civil society groups and well-resourced stakeholders. Small business owners and individual entrepreneurs face barriers including time constraints, lack of technical understanding of legislative language, and absence of accessible platforms for engagement.

Nyamu-Musembi (2002) discusses the concept of "participatory citizenship" and argues that effective participation requires not only formal opportunities but also substantive capacity among citizens to engage meaningfully. This includes access to information, understanding of

processes, and organizational capacity to articulate collective interests. Platforms like Bizi can contribute to building this capacity by making legislative information accessible and creating spaces for collective engagement.

2.6 Review of Existing Systems

2.6.1 Local Systems

Mzalendo (www.mzalendo.com**):** As Kenya's primary parliamentary monitoring platform, Mzalendo provides a foundation for understanding citizen expectations regarding legislative transparency. The platform offers MP profiles with attendance records, committee memberships, and parliamentary contributions. Bill tracking features allow users to view legislation at various stages of the legislative process. However, Mzalendo lacks interactive features for user feedback, voting on bills, or networking among users. The platform primarily serves as an information resource rather than a participatory tool (Mzalendo Trust, 2017).

KenyaLaw (www.kenyalaw.org**):** The National Council for Law Reporting operates Kenya Law, which provides access to legal resources including legislation, case law, and treaties. While comprehensive in its coverage of enacted laws, the platform is designed for legal professionals and does not cater specifically to the business community. There are no features for tracking bills in progress or facilitating public input (National Council for Law Reporting, 2020).

Huduma Kenya (www.hudumakenya.go.ke**):** The Huduma Kenya platform focuses on government service delivery rather than legislative participation. While it demonstrates the government's capacity to develop citizen-facing digital platforms, it does not address the need for legislative engagement or business networking.

2.6.2 International Systems

Congress.gov (United States): The official platform of the U.S. Congress provides comprehensive legislative information including bill text, status, sponsors, and related documents. The platform allows users to search and filter legislation, subscribe to updates, and access video of proceedings. However, it remains primarily informational without mechanisms for public voting or feedback (Library of Congress, 2021).

They Vote For You (Australia): This platform, developed by the OpenAustralia Foundation, tracks how members of the Australian Parliament vote and allows users to compare their own positions with those of their representatives (OpenAustralia Foundation, 2019). Users can indicate their stance on policy areas and see which politicians align with their views. This feature represents a step toward participatory engagement, though it focuses on retrospective voting analysis rather than input on pending legislation.

Decide Madrid (Spain): Madrid's participatory democracy platform enables citizens to propose, debate, and vote on municipal policies and spending priorities (Madrid City Council, 2017). The platform implements participatory budgeting and allows binding citizen votes on specific proposals that meet participation thresholds. While operating at the municipal level, Decide Madrid demonstrates the potential for digital platforms to enable genuine citizen influence on policy decisions.

Pol.is (Various Countries): Pol.is is an open-source platform for gathering and analyzing public opinion on complex issues (Small et al., 2021). The platform uses machine learning to identify opinion clusters and areas of consensus. It has been deployed in Taiwan for public consultations on technology policy and in other contexts for deliberative engagement. The approach offers insights for how Bizi might analyze and present aggregated user opinions on legislation.

2.7 Technical Considerations

2.7.1 Web Development Frameworks

The selection of appropriate web development frameworks is critical to the successful implementation of a platform like Bizi. Modern web development typically employs frameworks for both frontend (client-side) and backend (server-side) development.

Frontend Frameworks: React.js, developed by Facebook, has emerged as one of the most popular JavaScript libraries for building user interfaces (Facebook Inc., 2021). React's component-based architecture enables the development of reusable UI elements, which is beneficial for a platform with multiple interactive features like Bizi. Vue.js and Angular represent alternative frontend frameworks with different design philosophies and trade-offs. Vue.js offers a gentler learning curve and flexibility, while Angular provides a comprehensive framework with strong typing through TypeScript (Osmani, 2017).

Backend Frameworks: The backend of web applications handles business logic, database operations, and API endpoints. Node.js with Express.js offers a JavaScript-based backend solution that enables developers to use a single language across the stack (Tilkov & Vinoski, 2010). Django (Python) and Laravel (PHP) are full-featured frameworks that provide built-in functionality for authentication, database management, and security. Ruby on Rails follows a convention-over-configuration philosophy that accelerates development (Hansson, 2005). For Bizi, the choice of backend framework should consider factors including developer expertise, community support, scalability, and security features.

Full-Stack Considerations: The emergence of full-stack frameworks and platforms has simplified web development. The MERN stack (MongoDB, Express.js, React.js, Node.js) offers a JavaScript-based full-stack solution with strong community support (Subramanian, 2019). Alternatively,

traditional LAMP stacks (Linux, Apache, MySQL, PHP) remain widely used and benefit from extensive documentation and hosting support.

2.7.2 Database Management Systems

Database design and management are fundamental to applications that store user data, content, and interaction records. The choice between relational (SQL) and non-relational (NoSQL) databases depends on data structure requirements and scalability needs.

Relational Databases: MySQL and PostgreSQL are popular open-source relational database management systems (RDBMS). Relational databases enforce data integrity through schemas and relationships, making them suitable for applications with structured data and complex queries (Codd, 1970). For Bizi, user profiles, bills, votes, and discussion threads have clear relationships that map well to relational models. PostgreSQL offers advanced features including JSON support, full-text search, and robust transaction handling.

NoSQL Databases: MongoDB represents the document-oriented NoSQL approach, storing data in flexible JSON-like documents (MongoDB Inc., 2021). NoSQL databases offer advantages for applications with rapidly evolving data structures or high-volume, distributed workloads. However, for a platform like Bizi where data relationships (users, votes, bills, comments) are important, relational databases may offer advantages in query efficiency and data integrity.

Hybrid Approaches: Many modern applications employ hybrid database architectures, using relational databases for core transactional data and NoSQL solutions for specific use cases such as caching, session management, or full-text search (Sadalage & Fowler, 2012). Redis, an in-memory data structure store, is commonly used alongside primary databases for caching and real-time features.

2.7.3 Authentication and Security

Security is paramount for platforms that manage user data and facilitate civic participation. Several authentication and security mechanisms are relevant to the Bizi platform.

User Authentication: Modern authentication typically employs token-based systems such as JSON Web Tokens (JWT) rather than traditional session-based authentication (Jones et al., 2015). OAuth 2.0 and OpenID Connect provide standardized protocols for authentication and authorization, enabling features like social login while maintaining security (Hardt, 2012). For Bizi, a robust authentication system must balance security with user convenience.

Data Protection: The protection of user data is both an ethical obligation and, in many jurisdictions, a legal requirement. Kenya's Data Protection Act (2019) establishes principles for the processing of personal data, including consent, purpose limitation, and security safeguards.

The Bizi platform must implement appropriate technical measures including encryption, access controls, and audit logging to comply with these requirements.

Security Best Practices: The Open Web Application Security Project (OWASP) provides guidance on securing web applications, including protection against common vulnerabilities such as injection attacks, cross-site scripting (XSS), and cross-site request forgery (CSRF) (OWASP Foundation, 2021). Implementing these protections is essential for maintaining user trust and platform integrity.

2.7.4 System Architecture Approaches

The architectural design of web applications affects scalability, maintainability, and performance. Two prominent architectural approaches merit consideration.

Monolithic Architecture: Traditional web applications often follow a monolithic architecture where all functionality resides in a single codebase and deployment unit. This approach simplifies development and deployment for smaller applications but can become challenging to scale and maintain as complexity grows (Richards, 2015).

Microservices Architecture: Microservices decompose applications into small, independently deployable services that communicate through APIs (Newman, 2015). This approach offers advantages for scaling, as individual services can be scaled independently based on demand. For Bizi, a microservices approach might separate bill tracking, user management, voting, and networking into distinct services. However, microservices introduce operational complexity that may not be justified for an initial implementation.

Service-Oriented Considerations: For a platform like Bizi, a pragmatic approach might involve a modular monolith design that organizes code into distinct modules while maintaining a single deployment unit. This approach balances the organizational benefits of modularity with the operational simplicity of monolithic deployment, allowing for future migration to microservices if scaling demands require it.

2.8 Identified Research Gaps

The review of literature and existing systems reveals several significant gaps that the Bizi project aims to address:

Gap 1: Absence of Business-Centric Legislative Engagement Platforms in Kenya
While Kenya has platforms for general parliamentary monitoring (Mzalendo) and legal information (Kenya Law), no platform specifically targets the business community for legislative engagement. Business owners have unique needs regarding economic legislation, yet they lack a dedicated channel for accessing relevant information and expressing their views. Bizi

addresses this gap by creating a platform specifically designed for business owners, entrepreneurs, and economic enthusiasts.

Gap 2: Lack of Participatory Features in Existing Legislative Tracking Systems
The reviewed legislative tracking systems are predominantly informational, providing data on bills and legislators but not enabling users to express their opinions or vote on proposed legislation. This represents a missed opportunity for aggregating public sentiment and informing legislators about stakeholder views. Bizi introduces voting mechanisms that allow users to indicate their support, opposition, or neutrality on bills, creating a measurable expression of business community sentiment.

Gap 3: Disconnect Between Legislative Tracking and Professional Networking
Existing platforms treat legislative engagement and business networking as separate domains. However, for business owners, these activities are interconnected—legislative outcomes affect business operations, and professional networks can amplify policy influence. By integrating legislative tracking with networking features, Bizi creates a comprehensive platform that addresses multiple needs of its target users.

Gap 4: Limited Mechanisms for Translating Digital Participation into Policy Influence
Many e-participation platforms successfully collect citizen input but lack mechanisms for ensuring that input reaches and influences decision-makers. Bizi addresses this gap by designing features for aggregating user votes and generating reports that can be shared with parliamentary committees, legislators, and other stakeholders.

Gap 5: Insufficient Accessibility of Legislative Information for Non-Expert Users
Parliamentary information is often presented in technical language and formats that are inaccessible to average citizens. Business owners without legal or political expertise may struggle to understand the implications of proposed legislation. Bizi aims to present legislative information in accessible language, with summaries and analyses that highlight business implications.

2.9 Conclusion

This literature review has examined the theoretical foundations, existing platforms, and technical considerations relevant to the development of the Bizi platform. The review of civic technology and e-participation literature establishes the importance of digital tools for enhancing citizen engagement in democratic processes and provides frameworks for evaluating and designing participatory platforms. The examination of legislative tracking systems reveals a global trend toward parliamentary transparency but also highlights the predominantly informational nature of existing platforms and the absence of robust participatory features.

The review of business networking platforms demonstrates the value that professionals derive from online connections and provides insights into effective features for facilitating networking. The analysis of Kenya's public participation framework reveals strong constitutional foundations but significant implementation challenges that create opportunities for technology-enabled solutions.

The comparative analysis of existing systems confirms that no current platform adequately addresses the intersection of business-focused legislative engagement and professional networking in the Kenyan context. This gap, combined with the constitutional mandate for public participation and the significant economic contribution of the business community, justifies the development of the Bizi platform.

The technical review identifies appropriate technologies and architectural approaches for implementing a robust, secure, and scalable platform. These considerations will inform the system design decisions presented in subsequent chapters.

In summary, the Bizi platform addresses a genuine need that existing systems do not meet. By combining legislative tracking, user voting, and business networking in a single platform tailored to the Kenyan business community, Bizi has the potential to enhance both democratic participation and economic collaboration. The following chapters detail the methodology, design, and implementation of this platform.

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CHAPTER 3: METHODOLOGY

3.1 System Development Methodology

The selection of an appropriate system development methodology is critical to the successful delivery of the Bizi platform. This section examines the chosen methodology and justifies its suitability for the project's requirements, constraints, and objectives.

3.1.1 Overview of Development Methodologies

System development methodologies provide structured frameworks for planning, designing, building, testing, and deploying software systems. Several methodologies were considered for the Bizi project:

Waterfall Model: The traditional sequential approach where each phase must be completed before the next begins. While offering clear documentation and structured progression, Waterfall is inflexible to changing requirements and delays testing until late in the development cycle.

Agile Methodology: An iterative and incremental approach that emphasizes flexibility, collaboration, and rapid delivery of working software. Agile methodologies, including Scrum and Kanban, accommodate changing requirements and promote continuous feedback.

Rapid Application Development (RAD): A methodology focused on rapid prototyping and iterative delivery with minimal planning. RAD emphasizes user involvement and the quick production of working systems through component-based construction.

Spiral Model: A risk-driven approach that combines elements of design and prototyping in stages, with each iteration involving planning, risk analysis, engineering, and evaluation.

3.1.2 Selected Methodology: Agile with Scrum Framework

For the development of the Bizi platform, the **Agile methodology with the Scrum framework** has been selected. This choice is justified by several factors aligned with the project's characteristics:

Iterative Development: The Bizi platform comprises multiple interconnected modules including user management, bill tracking, voting systems, and networking features. Agile's iterative approach allows for the development and delivery of these modules incrementally, with each iteration producing a potentially shippable product increment.

Flexibility to Changing Requirements: As a civic-tech platform operating in a dynamic legislative environment, Bizi may require adjustments based on user feedback, changing parliamentary

procedures, or emerging technical requirements. Agile accommodates these changes without disrupting the entire development process.

User-Centric Development: Agile emphasizes continuous user involvement and feedback. For Bizi, which targets business owners and entrepreneurs, regular validation of features against user expectations is essential for adoption and success.

Time-Constrained Delivery: With a seven-week timeline for Minimum Viable Product (MVP) development, Agile's sprint-based approach enables focused delivery of priority features within defined timeboxes.

Risk Management: By delivering working software in short iterations, Agile allows for early identification and mitigation of technical and functional risks.

3.1.3 Scrum Framework Implementation

The Scrum framework structures the Agile approach through defined roles, events, and artifacts:

Roles:

- **Product Owner:** Responsible for defining features, prioritizing the product backlog, and ensuring the development team delivers value. In this project, this role is fulfilled by the student developer with guidance from the project supervisor.
- **Scrum Master:** Facilitates the Scrum process and removes impediments. This role is self-managed by the student developer.
- **Development Team:** Responsible for delivering potentially shippable increments. The student developer fulfills this role.

Events:

- **Sprints:** The seven-week development period is divided into weekly sprints, each focused on delivering specific features or components.
- **Sprint Planning:** At the beginning of each sprint, tasks are selected from the product backlog based on priority and capacity.
- **Daily Stand-ups:** Brief self-assessment sessions to review progress and identify blockers.
- **Sprint Review:** At the end of each sprint, completed work is demonstrated and evaluated.
- **Sprint Retrospective:** Reflection on the sprint process to identify improvements.

Artifacts:

- **Product Backlog:** A prioritized list of all features, enhancements, and fixes for the platform.
- **Sprint Backlog:** The subset of product backlog items selected for the current sprint.
- **Increment:** The sum of all completed product backlog items, representing a working version of the platform.

3.2 Requirements Gathering

Requirements gathering is a critical phase that determines the features and constraints of the system. This section describes the methods used to collect requirements and presents the functional and non-functional requirements for the Bizi platform.

3.2.1 Collection Method

Multiple approaches were employed to gather comprehensive requirements for the Bizi platform:

Stakeholder Interviews:
Direct interviews were conducted with potential users of the platform to understand their needs, challenges, and expectations. The following stakeholder groups were engaged:

- **Small Business Owners (5 participants):** Interviews focused on their awareness of parliamentary bills, challenges in accessing legislative information, and interest in a platform for legislative engagement. Key findings included limited awareness of proposed legislation, difficulty understanding legal language, and desire for simplified summaries of bills affecting their industries.
- **Entrepreneurs and Startup Founders (4 participants):** Discussions explored their networking needs, interest in policy advocacy, and preferences for platform features. Participants expressed strong interest in connecting with other entrepreneurs and collectively influencing policies affecting the startup ecosystem.
- **Parliamentary Affairs Expert (1 participant):** A discussion with a researcher at the Parliamentary Budget Office provided understanding of the legislative process, stages of bill consideration, and opportunities for public input.

Observation of Existing Platforms:
A systematic review of existing legislative tracking and business networking platforms was conducted to identify common features, best practices, and gaps. Platforms reviewed included

Mzalendo, Congress.gov, TheyWorkForYou, LinkedIn, and various civic-tech applications. This observation informed the feature set and user experience design for Bizi.

Document Analysis:
Relevant documents were analyzed to understand the legal and procedural context:

- Constitution of Kenya (2010), particularly provisions on public participation
- National Assembly and Senate Standing Orders
- Public Participation Act framework documents
- Sample parliamentary bills and their progression records

Survey Questionnaire:
A structured questionnaire was distributed to 30 business owners and entrepreneurs through business networks and social media channels. The questionnaire gathered quantitative data on:

- Awareness of parliamentary activities affecting business
- Current sources of legislative information
- Interest in various platform features
- Preferred modes of engagement and communication
- Technical capabilities (internet access, device usage)

Focus Group Discussion:
A focus group session with 8 participants representing diverse business sectors was conducted to validate initial requirements and prioritize features.

Table 3.1: Requirements Gathering Activities Summary

Method	Participants/Sources	Key Contributions
Stakeholder Interviews	12 individuals	User needs, pain points, feature expectations
Platform Observation	6 platforms reviewed	Best practices, feature benchmarking
Document Analysis	Constitutional and parliamentary documents	Legal requirements, process understanding
Survey Questionnaire	3 respondents	Quantitative validation, preference data
Focus Group	8 participants	Feature prioritization, requirement validation

3.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system. The following non-functional requirements have been identified for the Bizi platform:

NFR1: Performance

- The system shall load pages within 3 seconds under normal network conditions.
- The system shall support at least 500 concurrent users without degradation in performance.
- Database queries shall execute within 1 second for standard operations.

NFR2: Scalability

- The system architecture shall allow for horizontal scaling to accommodate user growth.
- The database design shall efficiently handle growth to at least 100,000 registered users and 10,000 bills.

NFR3: Availability

- The system shall maintain 99% uptime availability.
- Planned maintenance shall be scheduled during low-usage periods with advance notification to users.

NFR4: Security

- User passwords shall be encrypted using industry-standard hashing algorithms (bcrypt or Argon2).
- All data transmission shall be encrypted using TLS/SSL protocols.
- The system shall implement protection against common web vulnerabilities (SQL injection, XSS, CSRF).
- User sessions shall expire after 30 minutes of inactivity.

NFR5: Usability

- The user interface shall be intuitive and require no more than 10 minutes of orientation for new users.
- The platform shall be accessible on desktop and mobile devices through responsive design.
- The system shall comply with WCAG 2.1 Level AA accessibility guidelines.

NFR6: Reliability

- The system shall implement automated database backups at least once daily.
- The system shall gracefully handle errors and provide meaningful error messages to users.

NFR7: Compatibility

- The platform shall be compatible with modern web browsers including Chrome, Firefox, Safari, and Edge (latest two versions).
- The responsive design shall support screen sizes from 320px (mobile) to 1920px (desktop) and above.

NFR8: Maintainability

- The codebase shall follow consistent coding standards and be adequately documented.
- The system shall use modular architecture to facilitate updates and enhancements.

NFR9: Data Integrity

- The system shall ensure consistency of user votes and prevent duplicate voting on the same bill.
- All user actions affecting data shall be logged for audit purposes.

NFR10: Compliance

- The system shall comply with Kenya's Data Protection Act (2019) regarding personal data handling.
- The system shall provide mechanisms for users to access, correct, and delete their personal data.

3.2.3 Functional Requirements

Functional requirements specify the behaviors and functions the system must perform. The following functional requirements are organized by module:

User Management Module

ID	Requirement	Priority
FR1.1	The system shall allow users to register using email address and password	High
FR1.2	The system shall verify user email addresses through confirmation links	High
FR1.3	The system shall allow users to log in using registered credentials	High
FR1.4	The system shall allow users to reset forgotten passwords	High
FR1.5	The system shall allow users to create and edit profile information including name, business name, industry, and bio	High
FR1.6	The system shall allow users to upload profile pictures	Medium
FR1.7	The system shall allow users to specify business interests and industry categories for personalized content	High
FR1.8	The system shall allow users to manage notification preferences	Medium
FR1.9	The system shall allow users to deactivate or delete their accounts	Medium
FR1.10	The system shall support different user roles (regular user, admin)	High

Bill Tracking Module

ID	Requirement	Priority

FR2.1	The system shall display a list of current parliamentary bills	High
FR2.2	The system shall categorize bills by type (economic, taxation, trade, labor, etc.)	High
FR2.3	The system shall display detailed information for each bill including title, sponsor, summary, and full text link	High
FR2.4	The system shall track and display the current stage of each bill (First Reading, Committee Stage, etc.)	High
FR2.5	The system shall provide a visual timeline showing the progression of each bill	Medium
FR2.6	The system shall allow users to search bills by keyword, category, or sponsor	High
FR2.7	The system shall allow users to filter bills by stage, category, or date	High
FR2.8	The system shall allow users to follow/track specific bills	High
FR2.9	The system shall notify users of status changes to their followed bills	High
FR2.10	The system shall provide simplified summaries highlighting business implications of bills	Medium

Voting and Feedback Module

ID	Requirement	Priority
FR3.1	The system shall allow registered users to vote on bills (Support, Oppose, Neutral)	High
FR3.2	The system shall prevent users from voting multiple times on the same bill	High

FR3.3	The system shall allow users to change their vote on a bill	Medium
FR3.4	The system shall display aggregated voting results for each bill	High
FR3.5	The system shall display voting results as percentages and visual charts	Medium
FR3.6	The system shall allow users to add optional comments when voting	Medium
FR3.7	The system shall generate reports summarizing community sentiment on bills	Medium
FR3.8	The system shall allow filtering of voting data by industry category	Low
FR3.9	The system shall display the user's own voting history	Medium

Discussion Forum Module

ID	Requirement	Priority
FR4.1	The system shall provide discussion forums organized by topics/categories	High
FR4.2	The system shall allow users to create new discussion threads	High
FR4.3	The system shall allow users to reply to existing discussion threads	High
FR4.4	The system shall allow users to attach bills to discussion threads	Medium
FR4.5	The system shall allow users to like/upvote discussion posts	Medium
FR4.6	The system shall display discussion threads sorted by recency or popularity	Medium
FR4.7	The system shall allow users to search within discussions	Medium

FR4.8	The system shall allow users to report inappropriate content	High
FR4.9	The system shall notify users of replies to their posts	Medium

Networking Module

ID	Requirement	Priority
FR5.1	The system shall allow users to send connection requests to other users	High
FR5.2	The system shall allow users to accept or decline connection requests	High
FR5.3	The system shall display a user's connections list	High
FR5.4	The system shall allow connected users to send private messages	High
FR5.5	The system shall allow users to search for other users by name, business, or industry	High
FR5.6	The system shall recommend potential connections based on industry and interests	Medium
FR5.7	The system shall display user profiles with professional information	High
FR5.8	The system shall allow users to view other users' public activity (votes, discussions)	Medium
FR5.9	The system shall allow users to block other users	Medium

Notification Module

ID	Requirement	Priority
FR6.1	The system shall send email notifications for important events (bill updates, messages)	High

FR6.2	The system shall display in-app notifications	High
FR6.3	The system shall allow users to mark notifications as read	Medium
FR6.4	The system shall allow users to configure notification preferences	Medium

Administration Module

ID	Requirement	Priority
FR7.1	The system shall allow administrators to add and update bill information	High
FR7.2	The system shall allow administrators to manage user accounts	High
FR7.3	The system shall allow administrators to moderate discussion content	High
FR7.4	The system shall provide administrators with dashboard analytics	Medium
FR7.5	The system shall allow administrators to manage bill categories	Medium

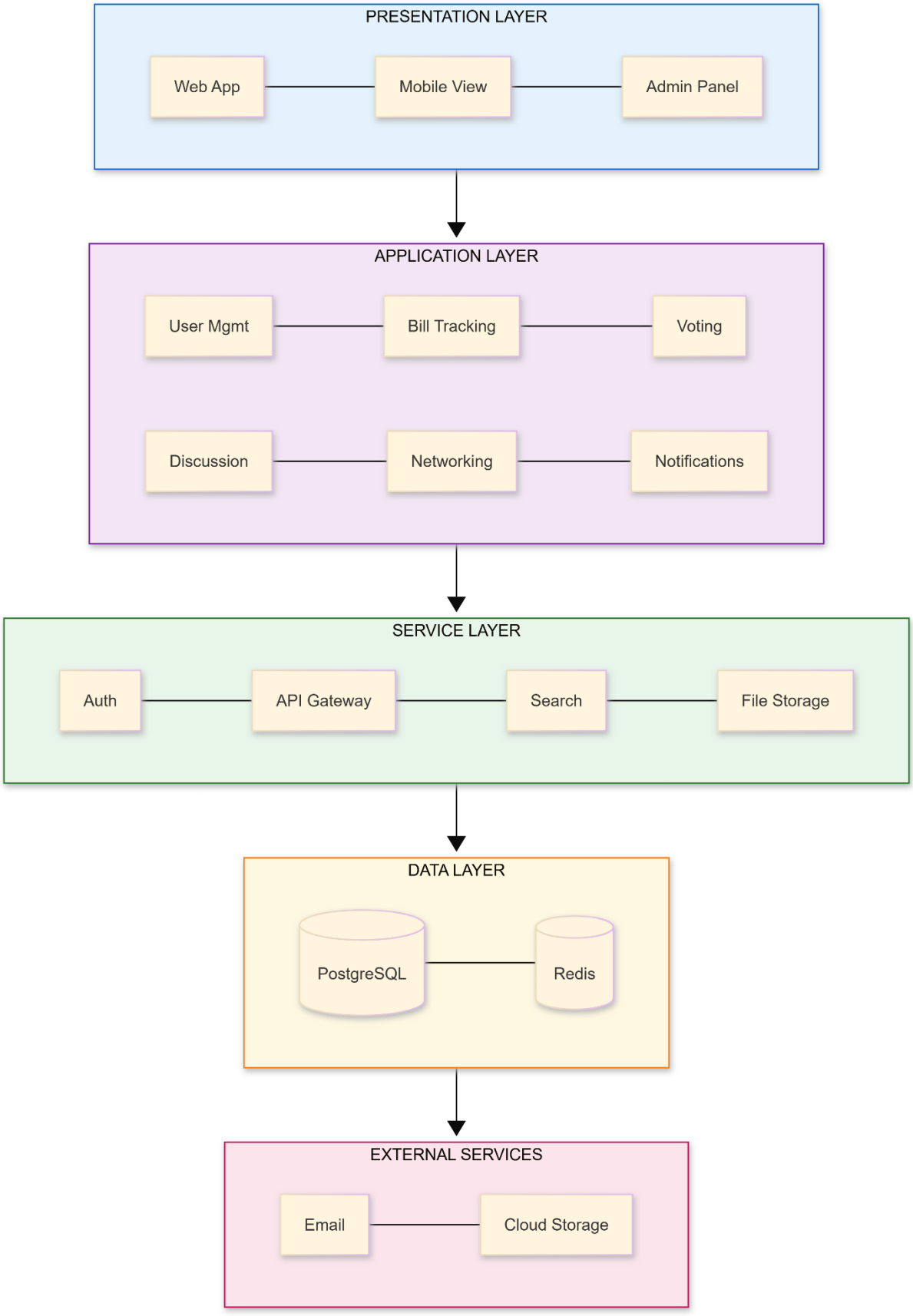
3.3 System Design

This section presents the architectural and structural design of the Bizi platform through various diagrams. Each diagram is explained to illustrate how the system components interact and how data flows through the system.

3.3.1 Block Diagram

The block diagram provides a high-level overview of the Bizi system architecture, showing the major components and their relationships. This diagram illustrates the system's structure without detailing internal processes.

Figure 3.2: Bizi Platform Block Diagram


PRESENTATION LAYER
Web App
Mobile View
Admin Panel
APPLICATION LAYER
User Mgmt
Bill Tracking
Voting
Discussion
Networking
Notifications
SERVICE LAYER
Auth
API Gateway
Search
File Storage
DATA LAYER
PostgreSQL
Redis
EXTERNAL SERVICES
Email
Cloud Storage

Explanation of the Block Diagram:

The Bizi platform follows a layered architecture pattern that promotes separation of concerns, maintainability, and scalability. The architecture comprises four primary layers:

Presentation Layer: This layer handles all user interface components. The Web App Interface provides the primary user experience through a browser-based application. The Mobile Responsive View ensures the interface adapts to various screen sizes for users accessing the platform via smartphones or tablets. The Admin Panel provides a separate interface for system administrators to manage content and users. Email Templates handle the formatting of notification emails sent to users.

Application Layer: This layer contains the core business logic of the platform, organized into functional modules. The User Management Module handles registration, authentication, and profile management. The Bill Tracking Module manages parliamentary bill data, categorization, and status tracking. The Voting Module processes user votes on bills and aggregates results. The Discussion Forum Module manages threads, posts, and replies. The Networking Module handles connections between users and private messaging. The Notification Module manages in-app and email notifications. The Admin Module provides administrative functionalities.

Service Layer: This layer provides shared services used across multiple modules. The Authentication Service manages user sessions and access control using JWT tokens. The API Gateway routes requests to appropriate modules and handles request validation. The Search Service provides full-text search capabilities across bills and discussions. The File Storage Service manages user uploads such as profile pictures.

Data Layer: This layer handles data persistence and caching. PostgreSQL serves as the primary relational database, storing all structured data including users, bills, votes, discussions, and connections. Redis provides caching for session data, frequently accessed content, and real-time features to improve performance.

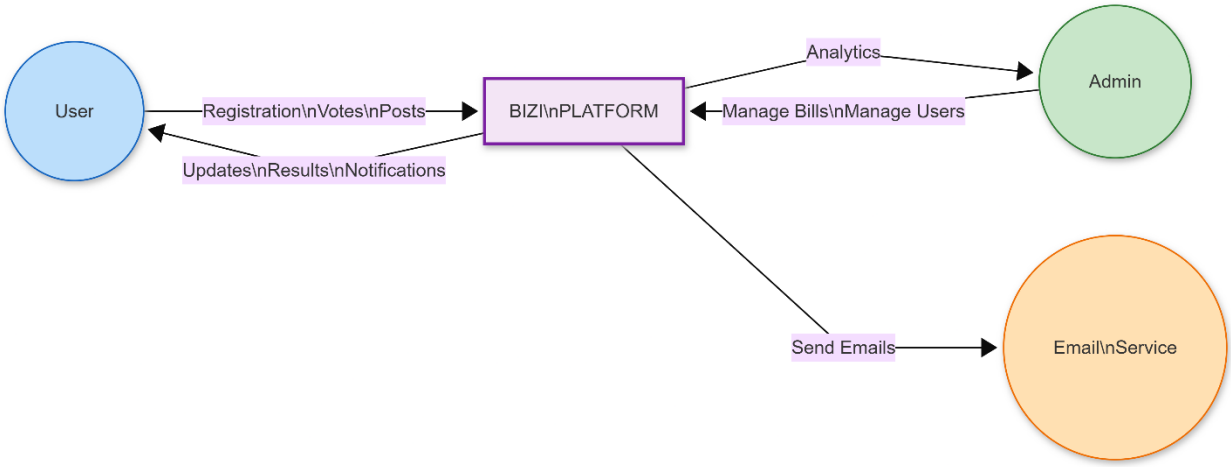
External Services: The platform integrates with external services including email delivery services (SMTP or third-party APIs like SendGrid), optional parliamentary data APIs where available, cloud storage for file uploads, and optional social authentication providers.

3.3.2 Data Flow Diagram

Data Flow Diagrams (DFDs) illustrate how data moves through the system, showing processes, data stores, data flows, and external entities. The following DFDs are presented at two levels: Context Diagram (Level 0) and Level 1 DFD.

Level 0: Context Diagram

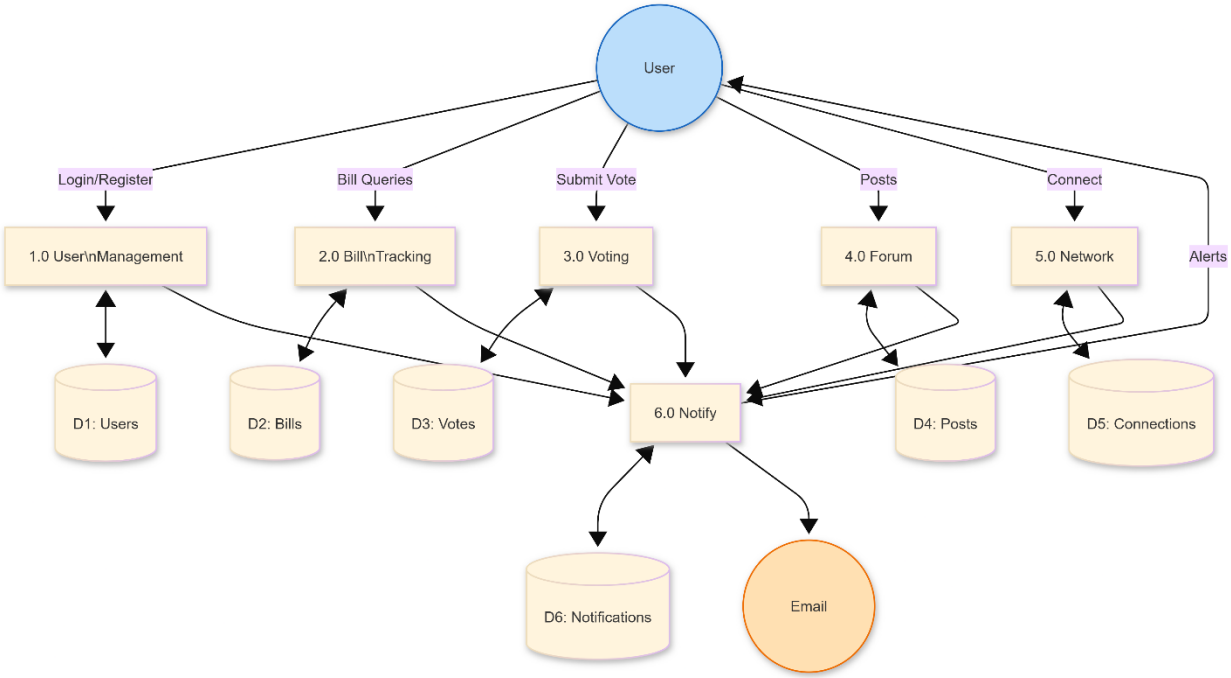
Figure 3.3: Context Diagram (Level 0 DFD)



Explanation: The context diagram shows the Bizi platform as a single process interacting with three external entities: Users (business owners and entrepreneurs who use the platform), Administrators (who manage content and users), and Email Service (external system for email delivery). Data flows between these entities and the platform are labeled to show the types of information exchanged.

Level 1: Data Flow Diagram

Figure 3.4: Level 1 Data Flow Diagram



Explanation of Level 1 DFD:

The Level 1 DFD decomposes the Bizi platform into six main processes, each handling specific functional areas:

Process 1.0 - User Management: Handles user registration, authentication, and profile management. It receives registration and login details from users, validates credentials against the User Database (D1), and returns authentication responses. User profile data is stored in and retrieved from the User Database.

Process 2.0 - Bill Tracking: Manages parliamentary bill data and user interactions with bills. It receives bill queries from users (searches, filters), retrieves relevant data from the Bill Database (D2), and returns bill information. User preferences from D1 inform personalized bill recommendations.

Process 3.0 - Voting: Processes user votes on bills. It receives vote submissions from authenticated users, validates against existing votes in the Vote Database (D3), stores new votes, and calculates aggregated results returned to users.

Process 4.0 - Discussion Forum: Manages discussion threads and posts. It receives new posts and replies from users, stores them in the Forum Database (D4), and retrieves discussion content based on user requests.

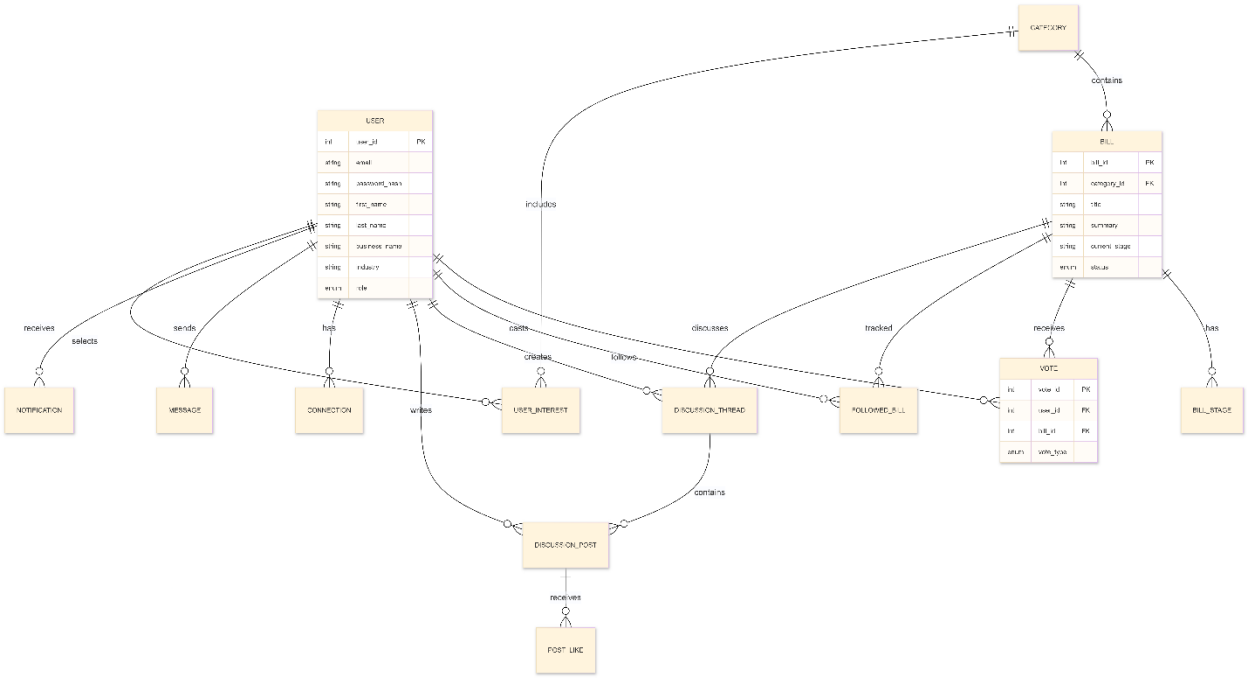
Process 5.0 - Networking: Handles connections between users and private messaging. It processes connection requests, manages the Connection Database (D5), and provides network updates and user recommendations.

Process 6.0 - Notification: Aggregates events from other processes and generates notifications. It stores notification records in the Notification Database (D6), delivers in-app notifications to users, and triggers email notifications through the external Email Service

3.3.3 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) models the database structure, showing entities, their attributes, and relationships between them.

Figure 3.5: Entity Relationship Diagram



Explanation of the ERD:

The Entity Relationship Diagram illustrates the database schema for the Bizi platform. The key entities and their relationships are explained below:

USER Entity: The central entity representing registered users of the platform. It stores authentication credentials (email, password_hash), personal information (first_name, last_name), business details (business_name, industry), and profile data (bio, profile_picture). The role attribute distinguishes between regular users and administrators.

CATEGORY Entity: Represents bill categories such as Taxation, Trade, Labor, Investment, etc. Categories are used to organize bills and user interests.

BILL Entity: Stores information about parliamentary bills including title, description, summary, sponsor, current_stage, and links to full text. Each bill belongs to one category (many-to-one relationship with CATEGORY).

BILL_STAGE Entity: Tracks the progression of bills through legislative stages. Each bill can have multiple stage records (one-to-many relationship), creating a history of the bill's journey through Parliament.

VOTE Entity: Records user votes on bills. Each vote is associated with one user and one bill (many-to-one relationships). The vote_type attribute stores the vote value (Support, Oppose, Neutral), and an optional comment allows users to explain their vote.

USER_INTEREST Entity: A junction table implementing the many-to-many relationship between users and categories. Users can specify interest in multiple categories for personalized content.

FOLLOWED_BILL Entity: A junction table tracking which users are following which bills, enabling notification of bill updates.

CONNECTION Entity: Represents networking connections between users. Both requester_id and receiver_id reference the USER entity (self-referencing many-to-many). The status attribute tracks the connection state (pending, accepted, declined).

MESSAGE Entity: Stores private messages between connected users. Both sender_id and receiver_id reference the USER entity.

DISCUSSION_THREAD Entity: Represents discussion forum threads. Each thread is created by a user, belongs to a category, and may optionally be associated with a specific bill.

DISCUSSION_POST Entity: Stores individual posts within discussion threads. Each post belongs to one thread and is authored by one user.

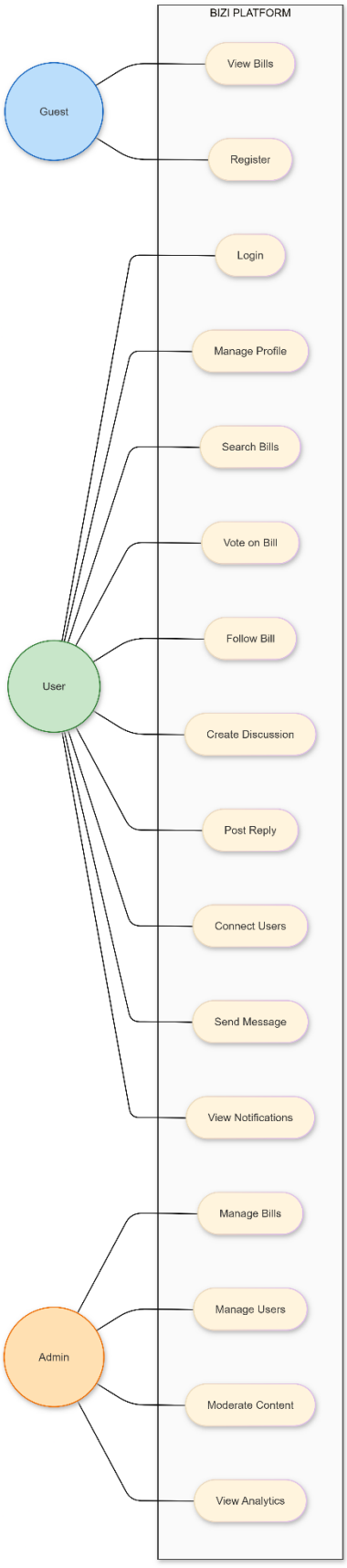
POST_LIKE Entity: Tracks user likes on discussion posts, implementing a many-to-many relationship between users and posts.

NOTIFICATION Entity: Stores notifications for users. The type attribute categorizes notifications (bill_update, message, connection_request, etc.), and reference_id links to the relevant entity (bill, message, connection).

3.3.4 Use Case Diagram

The Use Case Diagram illustrates the interactions between users (actors) and the system, showing the functionality available to different user types.

Figure 3.6: Use Case Diagram


BIZI PLATFORM
Guest
View Bills
Register
Login
Manage Profile
Search Bills
Vote on Bill
Follow Bill
User
Create Discussion
Post Reply
Connect Users
Send Message
View Notifications
Manage Bills
Manage Users
Admin
Moderate Content
View Analytics

Explanation of the Use Case Diagram:

The Use Case Diagram identifies three primary actors and their interactions with the Bizi platform:

Guest User (Unauthenticated):
Guest users can perform limited actions without registering:

- *View Public Bills:* Browse the list of parliamentary bills and their basic information.
- *View Public Discussions:* Read public discussion threads without participating.
- *Register Account:* Create a new user account, which includes email verification (<<include>> relationship).

Registered User (Authenticated):
After registration and login, users gain access to the full feature set:

- *Login:* Authenticate to access protected features.
- *Manage Profile:* Update personal and business information, including setting category interests (<<include>> relationship with Set Interests).
- *Search Bills / View Bill Details:* Search for bills and view comprehensive information including timeline (<<extend>> relationship with View Bill Timeline).
- *Follow Bill:* Subscribe to updates on specific bills.
- *Vote on Bill:* Express support, opposition, or neutrality on bills, optionally adding comments (<<extend>> relationship).
- *View Vote Results:* See aggregated voting results for bills.
- *Create Discussion Thread / Post in Discussion:* Participate in forum discussions.
- *Like Discussion Post:* Express appreciation for valuable contributions.
- *Search Users / View User Profile:* Discover and view other platform users.
- *Send Connection Request / Respond to Connection Request:* Build professional network.
- *Send Private Message:* Communicate privately with connections.
- *View Notifications / Manage Notification Settings:* Stay informed about relevant activities.
- *Report Content:* Flag inappropriate content for moderator review.

Admin (Administrator):
Administrators have elevated privileges for platform management:

- *Manage Bills:* Add new bills, edit bill information, and update bill stages (<<include>> relationships).
- *Manage Users:* View user accounts, suspend or delete users as necessary.
- *Moderate Content:* Review reported content, remove inappropriate posts, and lock discussions.
- *Manage Categories:* Create and modify bill and discussion categories.
- *View Analytics Dashboard:* Access platform usage statistics and insights.

3.4 Technical Requirements

This section outlines the technical requirements for developing and deploying the Bizi platform, including development tools, programming languages, frameworks, and hardware/software specifications.

3.4.1 Development Tools and Technologies

Frontend Development:

Component	Technology	Justification
JavaScript Framework	React.js 18.x	Component-based architecture, virtual DOM for performance, large ecosystem and community support
State Management	Redux Toolkit	Centralized state management for complex application state
UI Component Library	Tailwind CSS / Material-UI	Responsive design, consistent styling, accessibility features
HTTP Client	Axios	Promise-based HTTP client for API requests
Form Handling	React Hook Form	Efficient form validation and submission handling
Charts/Visualization	Chart.js / Recharts	Voting result visualization, analytics charts
Build Tool	Vite	Fast build times, hot module replacement

Backend Development:

Component	Technology	Justification
Runtime Environment	Node.js 18.x LTS	JavaScript runtime, non-blocking I/O, npm ecosystem
Web Framework	Express.js 4.x	Minimalist, flexible, widely adopted framework
Authentication	JSON Web Tokens (JWT)	Stateless authentication, scalable
Password Hashing	bcrypt	Industry-standard password hashing
Validation	Joi / express-validator	Request validation and sanitization
ORM	Prisma / Sequelize	Database abstraction, migrations, type safety
Email Service	Nodemailer with SendGrid	Reliable email delivery

Database:

Component	Technology	Justification
Primary Database	PostgreSQL 14.x	Robust RDBMS, ACID compliance, JSON support
Caching	Redis 7.x	Session storage, caching, real-time features
Database GUI	pgAdmin / DBeaver	Database administration and queries

Development Environment:

Component	Technology	Purpose
Version Control	Git with GitHub	Source code management, collaboration
IDE	Visual Studio Code	Code editing, debugging, extensions
API Testing	Postman / Insomnia	API development and testing
Container	Docker	Consistent development/deployment environments
Package Manager	npm / yarn	Dependency management

Testing:

Component	Technology	Purpose
Unit Testing	Jest	JavaScript testing framework
Integration Testing	Supertest	HTTP assertions for API testing
Frontend Testing	React Testing Library	Component testing
End-to-End Testing	Cypress (optional)	User journey testing

3.4.2 Software Requirements

Development Environment:

Software	Version	Purpose
Operating System	Windows 10/11, macOS, or Ubuntu 20.04+	Development machine OS
Node.js	18.x LTS	JavaScript runtime
npm	9.x	Package manager
PostgreSQL	14.x	Database
Redis	7.x	Caching
Git	2.x	Version control
Docker	Latest	Containerization

End User Requirements:

Component	Requirement
Web Browser	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Internet Connection	Minimum 1 Mbps broadband
JavaScript	Enabled
Cookies	Enabled
Screen Resolution	Minimum 320px width (mobile responsive)
Component	Requirement
Web Browser	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+

3.5 Budget

This section presents the estimated costs associated with the development and deployment of the Bizi platform MVP. The budget considers development tools, hosting services, and miscellaneous expenses.

3.5.1 Budget Breakdown

Table 3.2: Development Phase Budget

Item	Description	Quantity	Unit Cost (KES)	Total Cost (KES)
Development Tools				
Visual Studio Code	IDE (Free)	1	0	0
Git/GitHub	Version control (Free tier)	1	0	0
Postman	API testing (Free tier)	1	0	0
Figma	UI/UX design (Free tier)	1	0	0
Cloud Services (Development)				
DigitalOcean Droplet	Development server (Basic)	2 months	700	1,400
Managed PostgreSQL	Database hosting	2 months	1,500	3,000
Domain Name	.co.ke domain	1 year	1,200	1,200
Third-Party Services				
SendGrid	Email service (Free tier - 100/day)	1	0	0
Cloudinary	Image hosting (Free tier)	1	0	0
Hardware				
Development Laptop	Existing equipment	1	0	0
Internet Subscription	Monthly internet	2 months	3,500	7,000
Miscellaneous				
Electricity	Development period	2 months	2,000	4,000

Stationery	Documentation and notes	1	500	500
Transport	Stakeholder meetings	1	2,000	2,000
Contingency	Unexpected expenses (10%)	1	1,910	1,910
SUBTOTAL (Development)				**21,010**

Table 3.3: Deployment and Initial Operation Budget

Item	**Description**	**Quantity**	**Unit Cost (KES)**	**Total Cost (KES)**
Production Hosting				
DigitalOcean Droplet	Production server (Standard)	3 months	1,400	4,200
Managed PostgreSQL	Production database	3 months	2,500	7,500
Redis Cloud	Caching service	3 months	0 (Free tier)	0
SSL Certificate	Let's Encrypt (Free)	1	0	0
Third-Party Services				
SendGrid	Email service (Essentials plan)	3 months	1,900	5,700
Cloudinary	Image hosting (Plus plan)	3 months	0 (Free tier sufficient)	0
Maintenance				
Backup Storage	Automated backups	3 months	500	1,500
Monitoring	Uptime monitoring (Free tier)	3 months	0	0
SUBTOTAL (Deployment)				**18,900**

Table 3.4: Total Project Budget Summary

Phase	**Cost (KES)**
Development Phase	21,010
Deployment and Initial Operation	18,900
TOTAL PROJECT BUDGET	**39,910**

3.6 Timeline

The project follows a seven-week timeline for MVP development, structured according to the Agile Scrum methodology with weekly sprints. Each sprint focuses on specific deliverables aligned with project milestones.

3.6.1 Sprint Schedule

Sprint 1 (Week 1): Project Setup and User Management

- Development environment configuration
- Database schema design and implementation
- User registration and email verification
- User login and authentication (JWT)
- Password reset functionality
- Basic user profile management

Sprint 2 (Week 2): Bill Tracking Module

- Bill database model and seed data
- Bill listing page with pagination
- Bill detail page
- Bill categories and filtering
- Bill search functionality
- Bill following feature

Sprint 3 (Week 3): Voting and Feedback Module

- Vote submission functionality
- Vote validation (prevent duplicates)
- Vote aggregation and results display
- Vote commenting feature

- Voting history for users
- Results visualization (charts)

Sprint 4 (Week 4): Discussion Forum Module

- Discussion thread creation
- Discussion post/reply functionality
- Thread categorization
- Post liking feature
- Discussion search
- Basic moderation (report feature)

Sprint 5 (Week 5): Networking Module

- User search and discovery
- User profile viewing
- Connection request system
- Connection management (accept/decline)
- Private messaging
- Connection recommendations

Sprint 6 (Week 6): Notifications and Admin Module

- In-app notification system
- Email notification triggers
- Notification preferences
- Admin dashboard
- Bill management (CRUD)
- User management
- Content moderation tools

Sprint 7 (Week 7): Testing, Refinement, and Deployment

- Unit testing

- Integration testing
- User acceptance testing
- Bug fixes and refinements
- Performance optimization
- Production deployment
- Documentation completion

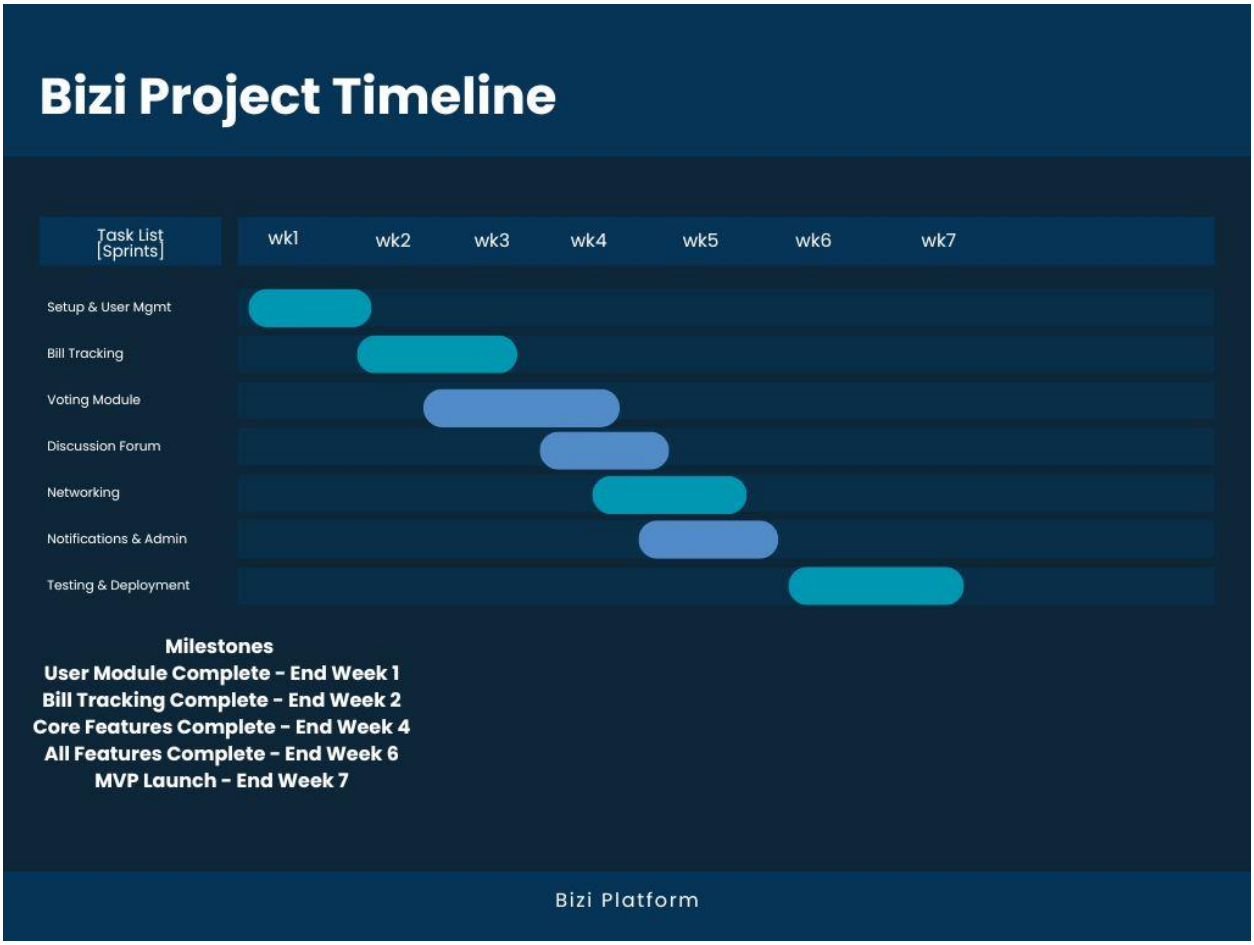


Figure 3.7: Project Gantt Chart

3.6.3 Timeline Considerations

Dependencies:

- Sprint 2 (Bill Tracking) depends on database setup from Sprint 1.
- Sprint 3 (Voting) depends on Bill and User modules.

- Sprint 5 (Networking) depends on User module completion.
- Sprint 6 (Notifications) depends on all other modules for event triggers.
- Sprint 7 (Testing) covers all previously developed modules.

Risk Mitigation:

- Buffer time is incorporated within each sprint for unexpected challenges.
- Features are prioritized to ensure core functionality is delivered first.
- Continuous integration allows for early detection of issues.

Flexibility:

- The Agile approach allows for reprioritization if certain features require more time.
- Lower priority features may be moved to post-MVP backlog if timeline pressure arises.